SCHEMATIC LOCUS OF A FREE FINITE GROUP QUOTIENT DETECTED BY AFFINE CHARTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1867-1875, immediately following the statement at 1854-1865.

Editorial note. Two cross-chapter pointers (Properties of Spaces, Decent Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- The quotient construction X/G. Properties of Spaces, Lemma spaces-properties-lemma-quotient, cited at proof lines 1867-1869. Supplies Y = X/G with X -> Y surjective etale; the groupoid X x_Y X = X x G.
- Quotient schematic locus for finite etale equivalence relations. Decent Spaces, Lemma decent-spaces-lemma-when-quotient-scheme-at-point, cited at proof lines 1873-1875. The imported terminal equivalence for quotient spaces by finite etale groupoids.

Source reference. Original reference 0BA1: focused-seeds/geometry/S9-sof-when-quotient-scheme-at-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 071S: focused-seeds/geometry/S9-sof-when-quotient-scheme-at-point/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S and let G be a finite group acting freely on X. Set $Y = X/G$ as in Properties of Spaces, Lemma 071S. For $y \in |Y|$ the following are equivalent*

(1) *y is in the schematic locus of Y, and*
(2) *there exists an affine open $U \subset X$ containing the preimage of y.*

Proof. It follows from the construction of $Y = X/G$ in Properties of Spaces, Lemma 071S that the morphism $X \to Y$ is surjective and étale. Of course we have $X \times_Y X = X \times G$ hence the morphism $X \to Y$ is even finite étale. It is also surjective. Thus the lemma follows from Decent Spaces, Lemma 0BA1. □

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